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Quadriceps EMG muscle activation during accurate soccer instep kicking

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Abstract
Six competitive soccer players were recruited to examine EMG activation in three quadriceps muscles during a kicking accuracy task. Participants performed three maximum instep place kicks of a stationary ball, 11 m perpendicular from the centre of the goal line towards targets (0.75 m^2) in the four corners of the goal. Surface EMG of the vastus lateralis, vastus medialis, and rectus femoris of the kicking leg was normalized and averaged across all participants to compare between muscles, targets, and the phase of the kick. Although no significant difference were observed between muscles or kick phases, kicks to the right targets produced significantly greater muscle activity than those towards the left targets ($P < 0.01$). In addition, kicks towards the top right target demonstrated significantly greater muscle activity than towards the top and bottom left ($P < 0.01$). Under accurate soccer shooting conditions, kicks aimed to the top right corner of the goal demonstrated a higher level of quadriceps muscle activation than those towards the other corners.

Keywords: *Electromyography, vastus lateralis, vastus medialis, rectus femoris, soccer, football*

Introduction

The two most common kicking actions used to score during the 1998 soccer World Cup were the instep kick and side-foot kick (Grant, Reilly, Williams, & Borrie, 1998). The instep kick, whereby the ball is hit with the dorsum of the foot, is the most powerful kicking action in soccer (Brophy, Backus, Pansy, Lyman, & Williams, 2007). Consequently, previous soccer-related biomechanical research has focused on the dynamics of the maximum instep kick, which corresponds to the penalty or "place" kick (Barfield, 1998; Brophy et al., 2007; Dörge et al., 1999; Dörge, Bull Andersen, Sørensen, & Simonsen, 2002; Levanon & Dapena, 1998; Luhtanen, 1988; Nunome, Asai, Ikegami, & Sakurai, 2002; Putnam & Dunn, 1987).

Previous research has examined the biomechanics of soccer movements in detail (Lees, 1996; Lees & Nolan, 1998), although only a limited number of studies have used electromyographic (EMG) analysis to observe muscle activation during the kick (De Proft, Clarys, Bollens, Cabri, & Dufour, 1988; Dörge et al., 1999; Kellis, Katis, & Gissis, 2004; McCrudden & Reilly, 1993; McDonald, 2002; Orchard, Walt, McIntosh, & Garlick, 2002). Although the whole body is known to influence muscular actions during the approach and kick, the quadriceps muscle group plays a vital role in the rapid extension of the knee joint in preparation for contact with the ball (Kellis & Katis, 2007). Previously reported normalized EMG values in maximum instep kicking range from 67% to 90% for the vastus medialis, 59% to 86% for the rectus femoris, and 64% to 102% for the vastus lateralis (De Proft et al., 1988; Dörge et al., 1999; Manolopoulos, Papadopoulous, & Kellis, 2006). This demonstrates the variability in previous research, which limits conclusions regarding the importance of each muscle during soccer kicking.

A maximum kick is not necessarily an accurate kick (Winter, 1990). Although studies on muscle activation patterns during accurate kicking are limited (Dicks & Kingman, 2005), previous research has indicated that muscle activation patterns may be more complex during accurate kicking to achieve a finer control of lower limb movement in order to make the necessary adjustments in muscular activity (Kellis & Katis, 2007). Accurate kicking is known to reduce the speed of the action compared with

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maximum kicking (Kellis & Katis, 2007). In addition, Kellis and Katis (2007) suggested that differences in muscle activity may occur during accurate kicking against a high or low target. Statistical analyses of right-footed penalty kicks taken in the UK Premier League from 2005 to 2008 (Lovejoy & Holmes, 2008) indicated that the largest percentage of kicks were taken to the bottom left corner of the goal, with 71% success compared with 100% success for kicks aimed at the top right corner and 93% success for the top left corner. However, the muscular demands of shooting to various corners of the goal are unknown.

Therefore, the aim of this study was to investigate EMG muscle activity of three quadriceps muscles during soccer kicking towards a high or a low target on the right and left sides of a goal. We hypothesized that there would be a significant difference in quadriceps muscle activity when shooting at high and low targets on left and right sides of the goal. We also hypothesized that there would be significant differences in peak muscular activity across each phase of the kick. Finally, we hypothesized that there would be no significant difference in the percentage time at which peak EMG activity occurred in each muscle when kicking towards the four targets.

Methods

Participants

Six male university-level competitive soccer players with 10 years or more experience volunteered for this study (mean $\pm$ *s*: age 21.2 ± 0.1 years, body mass 73.0 ± 7.2 kg, height 1.77 ± 0.06 m). All participants were right-footed, physically active, and had no history of lower limb, spinal or neurological injury. Participants provided signed written informed consent to partake and the institutional ethics committee approved the study.

Procedures

All participants performed a 5 min dynamic stretching warm-up. To generate reference EMG data for normalization purposes, each participant performed three dynamic squat-jumps immediately before the kicking protocol (Ball & Scurr, 2008). During the squat-jumps, the participants crouched with their feet shoulder-width apart, the back straight, and thighs parallel to the ground. After familiarization with the task, each participant performed maximum squat-jumps. Following this, each participant performed right-footed maximum velocity instep kicks of a stationary ball from a marked penalty spot. Three trials in four target conditions were undertaken. Targets (0.75 m^2) were positioned in the four corners of a standard size soccer goal, 11 m from the penalty spot. A ball of standard size and inflation was used to perform the trials. Participants were asked to hit the ball as hard and fast as possible, whilst maintaining accuracy towards the target area. Participants started their run-up from a marked point at an angle of 30° (Lees & Nolan, 1998). A rest period of 1 min between kicks was provided and only kicks placed in the target areas were counted.

During all conditions, EMG data were recorded at 1000 Hz and analysed using the Biometrics Datalog (UK) system. Electromyographic recordings were obtained from the vastus lateralis, vastus medialis, and rectus femoris of the right leg. The centre of each muscle belly was identified by palpation during voluntary isometric efforts. Electrode location and placement techniques were undertaken in accordance with SENIAM (Surface Electromyography for the Non-Invasive Assessment of Muscles) Group recommendations. Electrode location sites were prepared by shaving and cleansing, using an isopropyl alcohol swab (Medi-Swab, Seton Healthcare Group, UK) to reduce skin impedance ($\leq$10 kΩ). Active Ag/AgCl bipolar pre-amplified disc electrodes (SX230, Biometrics, UK; gain $\times$ 1000; input impedance $>$100 MΩ; common mode rejection ratio $>$96 dB; noise $<$1–2 μVRMS; bandwidth 20–450 Hz) with 1 cm separation distance were adhered in-line with the muscle fibres, using hypoallergenic adhesive tape (3M, UK). A passive reference electrode (R300, Biometrics, UK) was placed on the participant's olecranon process.

All electrodes were connected to the Datalog, which was securely attached to the participant's abdominal area during the task to minimize artifacts due to hardware movement and to enable the dynamic trials to be performed. The mass of the Datalog (0.5 kg) was assumed to have a negligible effect on kick performance. The Datalog used both a high-pass third-order filter (18 dB/octave; 20 Hz) to remove DC offsets due to membrane potential, and a low-pass filter for frequencies above 450 Hz. The electrodes contained an eighth-order elliptical filter ($-$60 dB at 550 Hz). An event marker (IS3, Biometrics, UK) determined foot–ball contact and ball–target contact. All EMG equipment was calibrated in accordance with the manufacturer's guidelines.

Data analysis

Electromyography data were uploaded to a Toshiba Laptop (Japan) and analysed using the Biometrics Datalog Analysis Package Software (Version 5.02, Biometrics, UK). The raw EMG signals (mV) were visually checked for artifacts. Raw EMG was then processed using root mean square (window length of

20 ms) and peak values were identified and expressed as a percentage of the peak root mean square EMG recorded during the squat-jumps. These percentage EMG values and the time at which they occurred (normalized to the percentage of total kick duration) were established during three phases of the kick: Phase 1 corresponded to the period when the participant started their movement (EMG recording initiated) until foot–ball contact (determined by the event marker); Phase 2 was defined as the period from foot–ball contact until ball–target contact (determined by the event marker); Phase 3 was the period from ball–target contact to the end of the movement.

Statistical analysis

Each participant's results were averaged across the three trials for each muscle, target area, and phase of the kick, then analysed using the statistical package SPSS (v.15). All data were normally distributed (Kolmogorov-Smirnoff and Shapiro-Wilk tests, $P > 0.05$). A repeated-measures analysis of variance (ANOVA) with two within-participant variables (target, with 4 levels and phase, with 3 levels) and one between-participant variable (muscle, with 3 levels) was applied to the normalized EMG data. Mauchley's test of sphericity showed that sphericity was not assumed and therefore a correctional factor was applied (Greenhouse-Geisser). *Post-hoc* paired *t*-tests with a Bonferroni adjustment ($P = 0.008$) were used to determine where significant differences lay. Then, three repeated-measures ANOVAs (one for each phase of the kick) with one within-participant variable (target, with 4 levels) and one between-participant variable (muscle, with 3 levels) were used to compare the time at which peak EMG activity occurred. Mauchley's test of sphericity showed that sphericity was assumed for these data. For all statistical tests, partial eta-squared (η^2) values were calculated to determine the effect size of the statistical analysis, with > 0.01 defining a small effect size, > 0.06 a moderate effect size, and > 0.14 a large effect size (Cohen, 1988). Statistical significance was set at $P < 0.05$ except where a Bonferroni adjustment was applied.

Results

Peak EMG values ranged from 57% in Phase 3 for the rectus femoris when kicking to the bottom left of the goal to 121% for the vastus lateralis in Phase 3 when kicking to the top right of the goal (Figure 1). Across all target areas, the vastus lateralis demonstrated a higher level of activation in Phase 1 compared with the other muscles. However, statistical analysis revealed no significant interaction effect in normalized EMG when accounting for target, muscle, and phase. When examining the main statistical effects, EMG activity displayed no significant difference across the three muscles of the quadriceps and no significant difference across each phase of the kick. A significant main effect on EMG activity was identified when kicking to different areas of the goal ($F_{2,13} = 4.34$, $P = 0.02$, partial $\eta^2 = 0.22$).

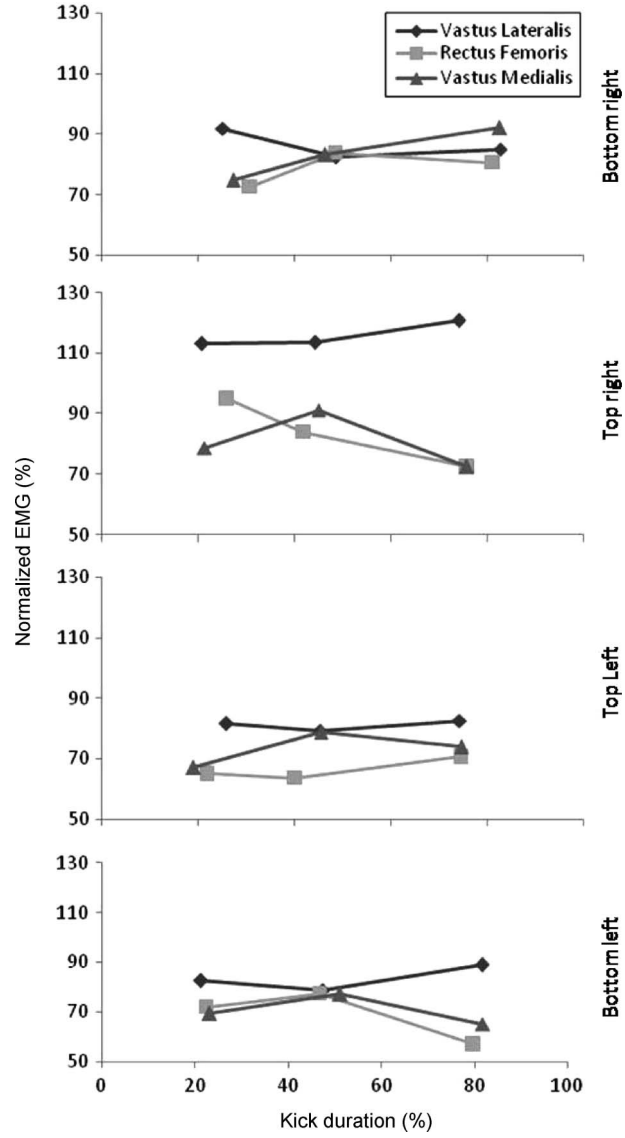

Figure 1. The average timing and amplitude of peak normalized EMG activity for three quadriceps muscles during three kicking phases to each target area during soccer instep kicking ($n = 6$).

Further analysis of the main effect of target showed greater overall muscle activity in the quadriceps when kicking to the top right of the goal. In addition, significantly greater quadriceps activity was required to kick the ball to the right side of the goal than the left side ($t_{111} = 4.59$, $P < 0.01$, partial $\eta^2 = 0.1$). There was no significant difference in quadriceps activity when kicking to the top or bottom of the goal ($t_{111} = 1.4$, $P = 0.16$, partial $\eta^2 = 0.1$).

The mean total duration of the kicks was 1.2 s ($s = 0.2$) with Phases 1, 2, and 3 lasting an average of 0.47 s ($s = 0.12$), 0.27 s ($s = 0.09$), and 0.5 s ($s = 0.19$), respectively. The instant at which peak EMG activity occurred in each phase showed no significant difference across targets and across muscles (Figure 1).

Discussion

The aim of this study was to investigate EMG muscle activity of three quadriceps muscles during soccer instep kicking towards four pre-defined targets. The results provided support for the first hypothesis, as there were significant differences in EMG activity when kicking towards different targets, with kicks directed towards the top right target demonstrating greater overall quadriceps muscle activation. The second hypothesis was rejected as there was no significant difference in peak quadriceps activation when compared across the phases of the kick. The third hypothesis was accepted, as the time at which peak EMG activity occurred showed no significant difference across the three muscles and across the four targets areas.

When averaged across all the targets, normalized EMG values of 77% for the rectus femoris, 89% for the vastus lateralis, and 83% for the vastus medialis were identified during the contact phase of the kick. These values are lower than previously reported at similar time points of an instep kick: Dörge et al. (1999) reported 93.7% for rectus femoris activity and 101.6% for vastus lateralis activity, while Brophy et al. (2007) reported 100% for the vastus medialis. However, as Dörge et al. (1999) and Brophy et al. (2007) analysed maximum instep kicks, whereas kicking accuracy was investigated in the present study, it is unsurprising that our values are lower.

Differences in muscular activation patterns were evident between targets. Kicks taken towards the right targets demonstrated significantly higher activation levels than kicks taken towards the left targets. Kicks towards the top right target demonstrated greater quadriceps activation than kicks towards the top and bottom left targets. Kicks towards the top right target demonstrated a greater activation of the vastus lateralis compared with the rectus femoris and vastus medialis, with peak vastus lateralis and vastus medialis activity occurring at similar times of each phase. This recruitment strategy may provide a positive contribution to the increased muscular activation during contact, stabilizing the knee joint during extension. This increase in muscular demand on the quadriceps when kicking towards the top right corner, coupled with the greater success rate of penalty kicks to this area of the goal (Lovejoy & Holmes, 2008), suggests that soccer penalty performance may be improved by training the quadriceps, with specific emphasis on the vastus lateralis and co-activation of the vastus lateralis and vastus medialis.

Despite no significant difference in the activation levels of the three quadriceps muscles, Figure 1 shows differences in quadriceps recruitment patterns throughout the kick. Peak rectus femoris activity was lower than peak vasti activity for most phases of the kick. Due to the biarticular nature of the rectus femoris, during a kicking action where the trunk is positioned over the legs, the rectus femoris is less dominant during knee extension as a consequence of hip flexion (Brophy et al., 2007). Figure 1 shows lower rectus femoris activity when kicking to the top of the goal, it is suggested that this may be a product of diminished rectus femoris responsibility as the participant changes the positioning of the trunk over the legs in Phase 2 in an attempt to accurately place the ball.

The issue of crosstalk in dynamic EMG assessment is recognized; however, previous research has quantified individual quadriceps activation during maximum instep kicking utilizing similar procedures (De Proft et al., 1988; Dörge et al., 1999; Manolopoulos et al., 2006). Other muscles around the hip, knee, and ankle may also play an important role when kicking towards different targets. Therefore, future research would benefit from recording muscle activity in the hip, knee, and ankle in comparison to the activity of the knee extensors assessed in this study to fully quantify the interactive effects of muscular activation on the outcome of penalty kicks towards differing targets. Future research may also benefit from synchronized kinematic and EMG data collection to identify changes in ball–foot contact when kicking at different targets. Furthermore, since all the participants in the present study were right-foot dominant and the results found greater quadriceps activation for these players when kicking towards the top right corner of the goal, future research should investigate quadriceps recruitment in left-footed soccer players. This study was a preliminary investigation into an under-reported area of soccer performance, the small sample size limits the generalizability of the conclusions, and given the variability of EMG measures further studies larger greater sample sizes are required.

In conclusion, the results of this study demonstrate a significant increase in EMG activation in the vastus lateralis, rectus femoris, and vastus medialis for kicks directed towards the top right target of the goal compared with kicks towards the bottom and left of the goal. This target condition demonstrated higher peak activation of the vastus lateralis and co-activation of the vastus medialis and vastus lateralis muscles throughout the kick. It is suggested that this may produce a rapid, but stabilized extension of the

knee joint compared with kicks directed towards the top left and the bottom corners of the goal. This study provides baseline data for further investigation into the changes in muscular recruitment patterns during various kicking situations, including specific target areas of the goal, across differing player skill levels, at varying states of fatigue, and after injury.

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